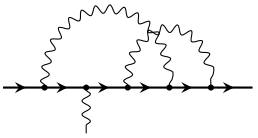
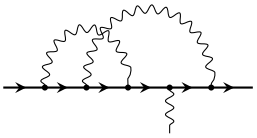
Diagrams evaluated by `sincqed`

Each diagram is labelled by the program entry that evaluates it, its Volkov string (external-photon vertex; photon ends, counting vertices along the electron line from 1) and, in brackets, the weight with which that entry already includes it. Entries of weight 2 include the mirror diagram, drawn next to its representative. Electron loops are drawn as circles (one-loop vacuum polarization), a circle with an internal photon (two-loop vacuum polarization), or a box (light-by-light).

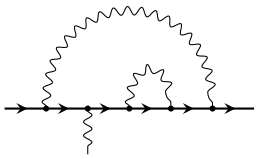
Two loops (7 diagrams)



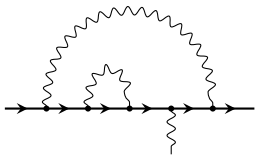
g2_00 2; 1-4, 3-5 [2]



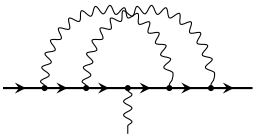
g2_00 mirror: 4; 1-3, 2-5



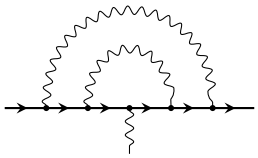
g2_01 2; 1-5, 3-4 [2]



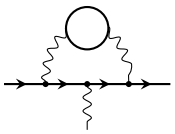
g2_01 mirror: 4; 1-5, 2-3



g2_02 3; 1-4, 2-5 [1]

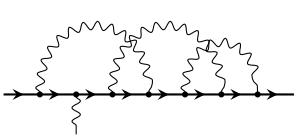


g2_03 3; 1-5, 2-4 [1]

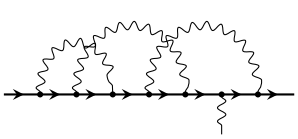


v2_00 one-loop VP [1]

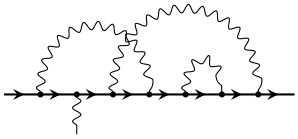
Three loops without lepton loops (50 diagrams, group `q3`)



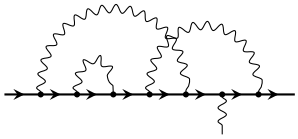
g3_00 2; 1-4, 3-6, 5-7 [2]



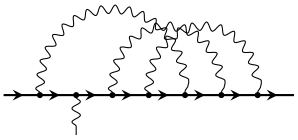
g3_00 mirror: 6; 1-3, 2-5, 4-7



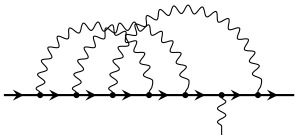
g3_01 2; 1-4, 3-7, 5-6 [2]



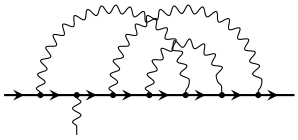
g3_01 mirror: 6; 1-5, 2-3, 4-7



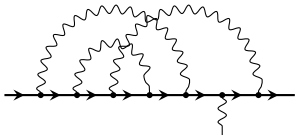
g3_02 2; 1-5, 3-6, 4-7 [2]



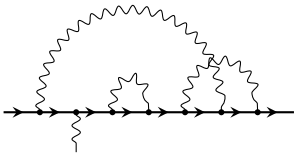
g3_02 mirror: 6; 1-4, 2-5, 3-7



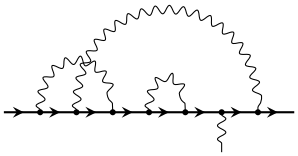
g3_03 2; 1-5, 3-7, 4-6 [2]



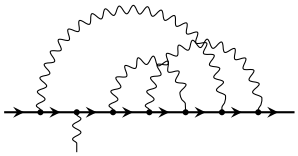
g3_03 mirror: 6; 1-5, 2-4, 3-7



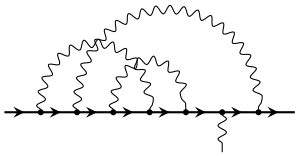
g3_04 2; 1-6, 3-4, 5-7 [2]



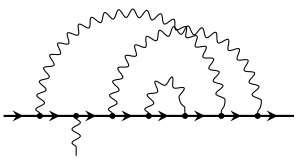
g3_04 mirror: 6; 1-3, 2-7, 4-5



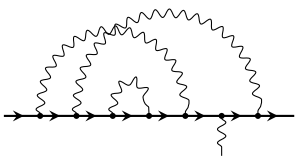
g3_05 2; 1-6, 3-5, 4-7 [2]



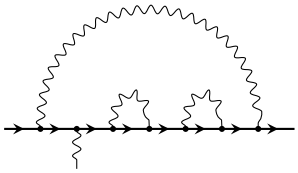
g3_05 mirror: 6; 1-4, 2-7, 3-5



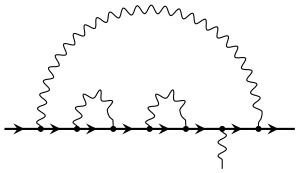
g3_06 2; 1-6, 3-7, 4-5 [2]



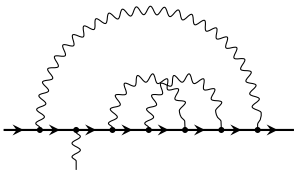
g3_06 mirror: 6; 1-5, 2-7, 3-4



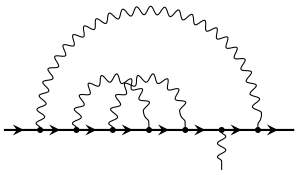
g3_07 2; 1-7, 3-4, 5-6 [2]



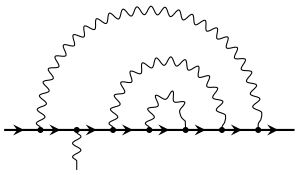
g3_07 mirror: 6; 1-7, 2-3, 4-5



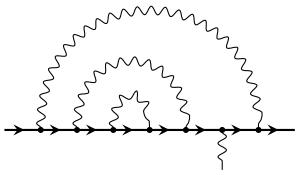
g3_08 2; 1-7, 3-5, 4-6 [2]



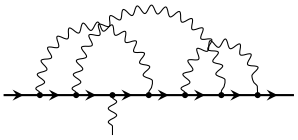
g3_08 mirror: 6; 1-7, 2-4, 3-5



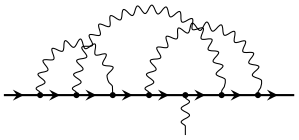
g3_09 2; 1-7, 3-6, 4-5 [2]



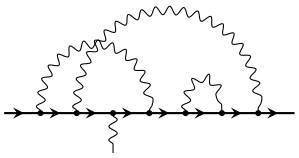
g3_09 mirror: 6; 1-7, 2-5, 3-4



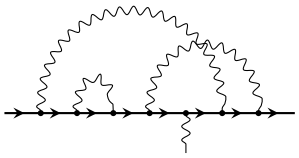
g3_10 3; 1-4, 2-6, 5-7 [2]



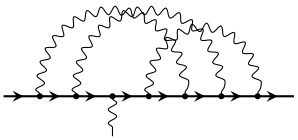
g3_10 mirror: 5; 1-3, 2-6, 4-7



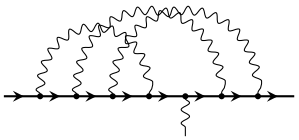
g3_11 3; 1-4, 2-7, 5-6 [2]



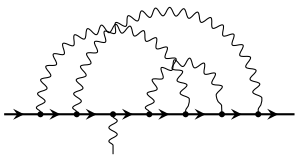
g3_11 mirror: 5; 1-6, 2-3, 4-7



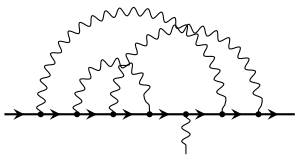
g3_12 3; 1-5, 2-6, 4-7 [2]



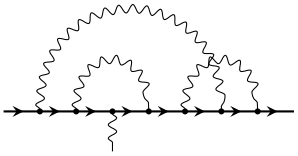
g3_12 mirror: 5; 1-4, 2-6, 3-7



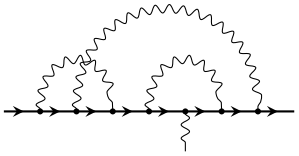
g3_13 3; 1-5, 2-7, 4-6 [2]



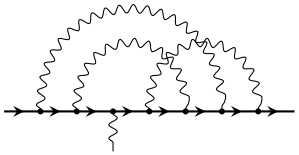
g3_13 mirror: 5; 1-6, 2-4, 3-7



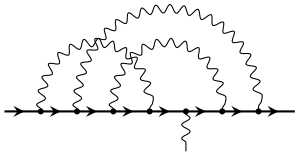
g3_14 3; 1-6, 2-4, 5-7 [2]



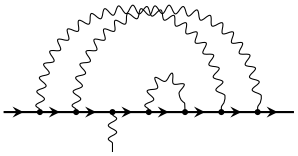
g3_14 mirror: 5; 1-3, 2-7, 4-6



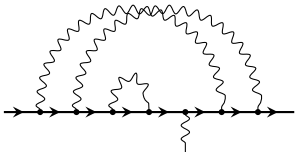
g3_15 3; 1-6, 2-5, 4-7 [2]



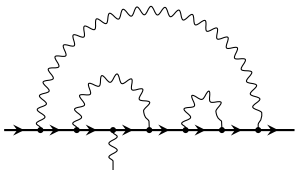
g3_15 mirror: 5; 1-4, 2-7, 3-6



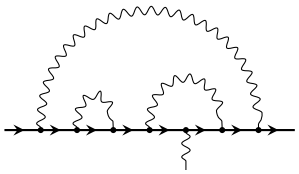
g3_16 3; 1-6, 2-7, 4-5 [2]



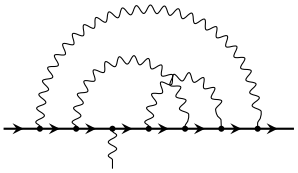
g3_16 mirror: 5; 1-6, 2-7, 3-4



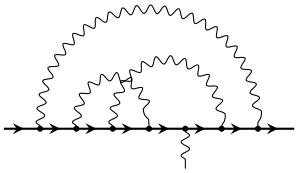
g3_17 3; 1-7, 2-4, 5-6 [2]



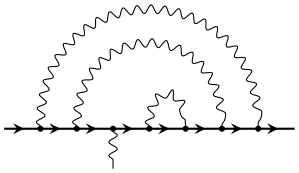
g3_17 mirror: 5; 1-7, 2-3, 4-6



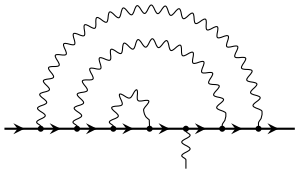
g3_18 3; 1-7, 2-5, 4-6 [2]



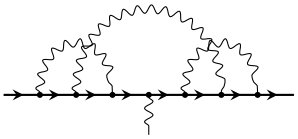
g3_18 mirror: 5; 1-7, 2-4, 3-6



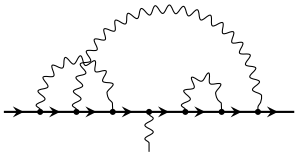
g3_19 3; 1-7, 2-6, 4-5 [2]



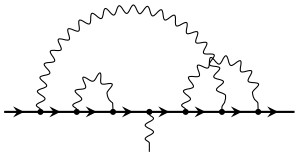
g3_19 mirror: 5; 1-7, 2-6, 3-4



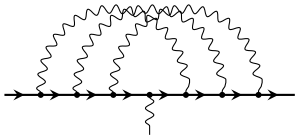
g3_20 4; 1-3, 2-6, 5-7 [1]



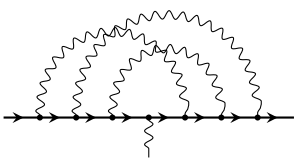
g3_21 4; 1-3, 2-7, 5-6 [2]



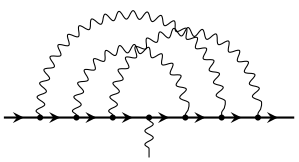
g3_21 mirror: 4; 1-6, 2-3, 5-7



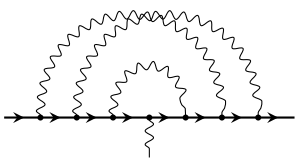
g3_22 4; 1-5, 2-6, 3-7 [1]



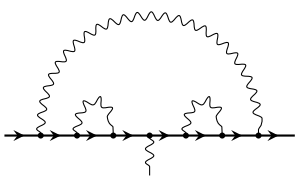
g3_23 4; 1-5, 2-7, 3-6 [2]



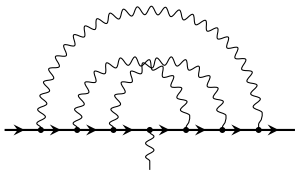
g3_23 mirror: 4; 1-6, 2-5, 3-7



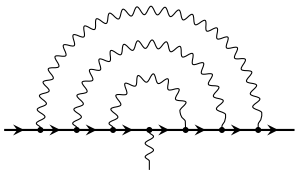
g3_24 4; 1-6, 2-7, 3-5 [1]



g3_25 4; 1-7, 2-3, 5-6 [1]

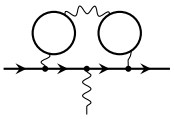


g3_26 4; 1-7, 2-5, 3-6 [1]

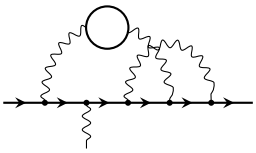


g3_27 4; 1-7, 2-6, 3-5 [1]

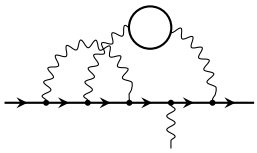
Three loops with vacuum-polarization insertions (16 diagrams, group vp3)



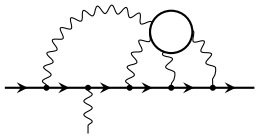
v3_00 two one-loop VPs [1]



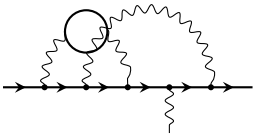
v3_01 2; 1-4, 3-5, VP on 1–4 [2]



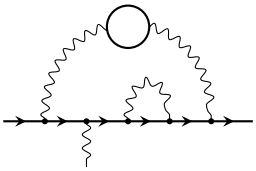
v3_01 mirror: 4; 1-3, 2-5



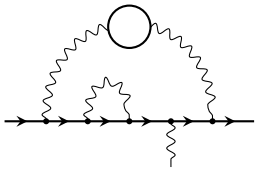
v3_02 2; 1-4, 3-5, VP on 3–5 [2]



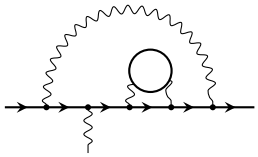
v3_02 mirror: 4; 1-3, 2-5



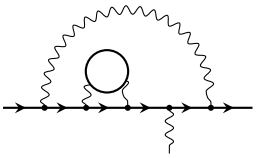
v3_03 2; 1-5, 3-4, VP on 1–5 [2]



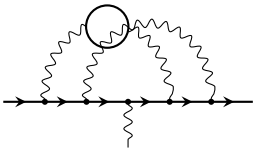
v3_03 mirror: 4; 1-5, 2-3



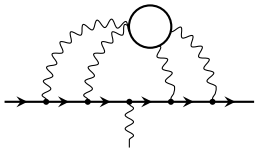
v3_04 2; 1-5, 3-4, VP on 3–4 [2]



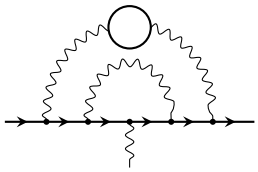
v3_04 mirror: 4; 1-5, 2-3



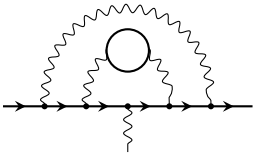
v3_05 3; 1-4, 2-5, VP on 1–4 [2]



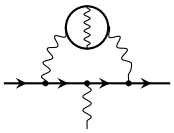
v3_05 mirror: VP on 2–5



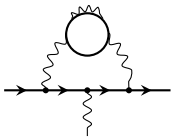
v3_06 3; 1-5, 2-4, VP on 1–5 [1]



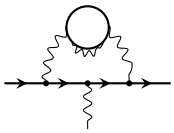
v3_07 3; 1-5, 2-4, VP on 2–4 [1]



v3_p4Vm + v3_p4Vc two-loop VP, vertex type [1]



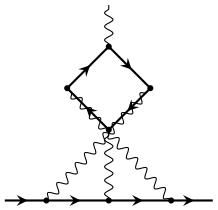
v3_p4Sm + v3_p4Sc two-loop VP, self-energy type [2]



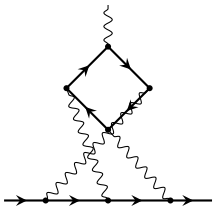
(included in v3_p4Sm, v3_p4Sc)

Three-loop light-by-light (6 diagrams, entry g3_lbl, all orderings summed pointwise)

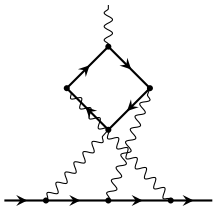
Photons 1, 2, 3 attach left to right on the open line; the ordering gives the loop vertex (1 right, 2 bottom, 3 left of the box) of each.



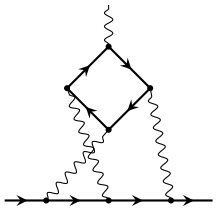
g3_lbl ordering 123



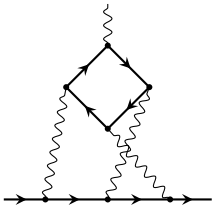
g3_lbl ordering 132



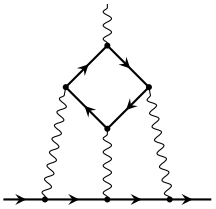
g3_lbl ordering 213



g3_lbl ordering 231



`g3_lbl` ordering 312



`g3_lbl` ordering 321